

Completing the square



1 Halve the coefficient of x and then square it.

2

a No

b 9

3 $\frac{9}{4}$

4

a 25

b 1

c 8

d $\frac{1}{4}$

e $\left(\frac{m}{2}\right)^2$

f $\frac{4}{25}$

5

a $x^2 + 4x + 4 = (x + 2)^2$

b $x^2 - 5x + \frac{25}{4} = \left(x - \frac{5}{2}\right)^2$

c $x^2 - \frac{7}{4}x + \frac{49}{64} = \left(x - \frac{7}{8}\right)^2$

6

a $(x + 6)^2 - 36$

b $(x - 6)^2 - 36$

c $(x + 1)^2 + 1$

d $(x - 7)^2 + 7$

e $(x - 3)^2 - 4$

f $\left(x + \frac{3}{2}\right)^2 + \frac{15}{4}$

g $\left(x - \frac{7}{2}\right)^2 + \frac{7}{4}$

7

a $y = (x + 3)(x + 1)$

b $y = (x - 2)(x - 4)$

c $y = (x + 7)(x + 10)$

d $y = (x - 4)(x - 7)$

e $y = (x + 53)(x + 3)$

f $y = (x - 3)(x - 21)$

8 $y = (x + 10)(x + 2)$

9

a $y = (x - 3 + \sqrt{2})(x - 3 - \sqrt{2})$

b $y = (x + 4 + \sqrt{2})(x + 4 - \sqrt{2})$

10

a In step 4, she did not factor the expression, but instead wrote it as the difference of two squares.

b $y = (x - 3)(x - 35)$



Non-monic completing the square

11 $4 \left(\left(x - \frac{9}{2} \right)^2 - \frac{1}{2} \right)$

12 a $3 \left(x + \frac{3}{2} \right)^2 + \frac{5}{4}$

b $4 \left(x - \frac{11}{8} \right)^2 - \frac{9}{16}$

13 a $y = (2x + 23)(x + 1)$

b $y = 4(x + 9)(x + 2)$

Quadratic equations

- 14 a No b Yes c No d No
e Yes

- 15 a $x = -4, -8$ b $x = \frac{3}{2}, \frac{1}{2}$ c $x = 5, 1$ d $x = \pm 2\sqrt{5} - 2$
e $x = -5 \pm \sqrt{17}$ f $x = -9 \pm \frac{\sqrt{30}}{2}$

- 16 a $x = -2, -16$ b $x = 4, 2$ c $x = 8, 1$ d $x = 8, -2$
e $x = -1, -\frac{7}{4}$ f $x = 1 \pm \sqrt{33}$

- 17 a $x = -12 \pm \sqrt{139}$ b $x = \frac{-11 \pm \sqrt{101}}{2}$
c $x = \pm 2\sqrt{3} - 4$ d $x = \frac{7}{2} \pm \frac{\sqrt{17}}{2}$
e $x = -\frac{1}{6} \pm \frac{\sqrt{109}}{6}$ f $x = \frac{-11}{2} \pm \frac{\sqrt{2965}}{10}$

Applications



19 $8, -8$

20 $x = 12$